

NAME:TANKISO MASOEBE

OPERATING SYSTEM(ADVANCED)WK1

1.OS Scenario: Memory Management

Memory management is an important function of an operating system because it controls how a computer uses its available RAM. For example, when a student opens a web browser, Microsoft Word, and a music application at the same time, the operating system must allocate enough memory to each program. It keeps track of which areas of memory are being used and which are free. If the available RAM becomes low, the operating system can use virtual memory, where some data is temporarily moved from RAM to storage. This allows applications to continue running even when physical memory is limited. The operating system also prevents one program from accessing another program's memory, improving security and stability. Effective memory management ensures that programs run smoothly, reduces crashes, and allows multiple applications to operate at the same time without interfering with one another.

2.Virtual Memory

Virtual memory is an operating system concept that allows a computer to use part of its storage drive as if it were additional RAM. It enables programs to run even when the physical memory is almost full. The operating system divides memory into pages and moves less-used pages between RAM and storage when necessary. This helps improve memory management and allows multiple applications to run at the same time. Virtual memory also provides each program with its own protected memory space, improving system security and stability. Although storage is slower than RAM, virtual memory helps prevent programs from stopping because of insufficient physical memory.

3.Tool Practice Reflection

For this activity, I used VirtualBox to simulate an operating system environment and practice process monitoring. I observed how different processes run in the system and how they use resources such as CPU and memory. This activity helped me understand that an operating system manages processes and ensures that applications receive the resources they need. I also learned that process monitoring can help identify programs that use too many system resources or are not responding. Using VirtualBox was useful because it provided a safe environment for practicing without affecting the main computer. Overall, the activity improved my understanding of process management and OS resource allocation.

Tool Practice,.

1. Open **VirtualBox** and start your virtual machine.
2. Open the operating system inside the virtual machine.
3. Open **Task Manager** (Ctrl + Shift + Esc on Windows).
4. Go to the **Processes** tab.
5. Observe running applications such as Chrome, File Explorer, and system processes.
6. Check their **CPU, Memory, Disk, and Network** usage.
7. Select a process and observe how its resource usage changes.
8. Take a **screenshot** of Task Manager showing the processes and resource usage.
9. Use the screenshot as evidence for your assignment.

What the activity demonstrates:

The operating system monitors processes and manages resources such as CPU and memory to keep the computer running efficiently.

4.OS Process States Explanation

An operating system manages processes through different states during their execution. A new process first enters the **New** state when it is created. It then moves to the **Ready** state, where it waits for the CPU to become available. When the CPU is assigned, the process enters the **Running** state and executes its instructions. If the process needs input or another resource, it moves to the **Waiting** state. After the required resource becomes available, it returns to the Ready state. Finally, when the process completes its task, it moves to the **Terminated** state. This process helps the operating system manage CPU resources efficiently.

OS PROCESS STATES DIAGRAM

The operating system manages a process through different states from creation to completion.

